THE SCHEME THEORETIC IMAGE OF A QUASI-COMPACT MORPHISM AFTER AN ADMISSIBLE BLOWUP IS THE STRICT TRANSFORM OF THE ORIGINAL IMAGE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Canonical location of the counted proposition: Stacks tag 0F3Y (sources/stacks-flat/pinned-original-source.tex line 9728).

Allowed standard prerequisites (editorial).

- Scheme theoretic images of quasi-compact morphisms agree with the images of the restrictions to open subschemes. Morphisms, Lemma labelled morphisms-lemma-quasi-compact-scheme-theoretic-image (Stacks tag 01R8), invoked at proof lines 9750-9752. Supplies that Z cap (U x_S Y) = Z" cap (U x_S Y) once both V -> X x_S Y and V -> X' x_S Y are quasi-compact.
- Quasi-compactness is preserved by base change. Schemes, Lemma labelled schemes-lemma-quasi-compact-permanence (Stacks tag 03GI), invoked at proof lines 9746-9749. Supplies that V -> X' x_S Y stays quasi-compact, which is required for the previous input.
- Strict transforms of closed subschemes along an admissible blowup and the absence of sections supported on the exceptional divisor. Divisors, the definition and elementary properties of strict transforms used in the statement. Supplies the identification of Z' and the property of its structure sheaf used at proof lines 9754-9758.

Source reference. Original lemma 03GI: sources/stacks-chow/tags.txt, source lines 4529–4529. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9748 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01R8: sources/stacks-chow/tags.txt, source lines 2293–2293. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9751 of the counted proof. It is

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Lemma 0.1. *Let $X \to S$ and $Y \to S$ be morphisms of schemes. Let $U \subset X$ be an open subscheme. Let $V \to X \times_S Y$ be a quasi-compact morphism whose composition with the first projection maps into U. Let $Z \subset X \times_S Y$ be the scheme theoretic image of $V \to X \times_S Y$. Let $X' \to X$ be a U-admissible blowup. Then the scheme theoretic image of $V \to X' \times_S Y$ is the strict transform of Z with respect to the blowing up.*

Proof. Denote $Z' \to Z$ the strict transform. The morphism $Z' \to X'$ induces a morphism $Z' \to X' \times_S Y$ which is a closed immersion (as Z' is a closed subscheme of $X' \times_X Z$ by definition). Thus to finish the proof it suffices to show that the scheme theoretic image Z'' of $V \to Z'$ is Z'. Observe that $Z'' \subset Z'$ is a closed subscheme such that $V \to Z'$ factors through Z''. Since both $V \to X \times_S Y$ and $V \to X' \times_S Y$ are quasi-compact (for the latter this follows from Schemes, Lemma 03GI and the fact that $X' \times_S Y \to X \times_S Y$ is separated as a base change of a proper morphism), by Morphisms, Lemma 01R8 we see that $Z \cap (U \times_S Y) = Z'' \cap (U \times_S Y)$. Thus the inclusion morphism $Z'' \to Z'$ is an isomorphism away from the exceptional divisor E of $Z' \to Z$. However, the structure sheaf of Z' does not have any nonzero sections supported on E (by definition of strict transforms) and we conclude that the surjection $\mathcal{O}_{Z'} \to \mathcal{O}_{Z''}$ must be an isomorphism. □